ECE 310 Fall 2026
Lecture 14: Discrete-time Fourier Transform Properties

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Learning Objectives

After this lecture, you should be able to:

- Explain the relationship between the z-transform and the DTFT.
- Apply key properties of the DTFT to compute transform pairs and analyze signals in both the time-domain and frequency-domain.



Last Time: Discrete-time Fourier Transform

Discrete-time Fourier Transform

The discrete-time Fourier transform (DTFT) of a discrete-time signal $x[n]$ is denoted as $X_d(\omega)$: $x[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega)$.

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X_d(\omega) e^{j\omega n} d\omega$$

The DTFT of the signal $x[n]$ exists if

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty.$$



Example: Computing the DTFT

Compute the DTFT of the following sequences.

$$x[n] = \{1, \underset{\uparrow}{0}, 0, -1\}$$

$$x[n] = a^n u[n],\ a \in \mathbb{C},\ |a| < 1$$

$$x[n] = \{\underset{\uparrow}{1}, 2, 3\}$$



How are the z-transform and DTFT related?

Thus far, we have seen:

- The z-transform shows how much each geometric series z^n, $z \in \mathbb{C}$ contributes to a discrete-time signal.
- The DTFT shows how much each periodic complex exponential $e^{j\omega_0 n}$, $\omega_0 \in \mathbb{R}$, contributes to a discrete-time signal.



Concept Check: z-transform and DTFT

Consider the following z-transforms:

$$X_1(z) = \frac{1 - z^{-1}}{1 - \frac{1}{2}z^{-1}}, \ |z| > \frac{1}{2}$$

$$X_2(z) = \frac{1 + z^{-1}}{1 - 2z^{-1} - 3z^{-2}}, \ |z| > 2$$

Determine the DTFT of each signal $X_1(\omega)$ and $X_2(\omega)$, if they each exist.

1. a) Does not exist b) $X_1(\omega) = \dfrac{1 - e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega}}$ c) $X_1(\omega) = \dfrac{1 - j\omega}{1 - \frac{1}{2}j\omega}$

2. a) Does not exist b) $X_2(\omega) = \dfrac{1 + e^{-j\omega}}{1 - 2e^{-j\omega} - 3e^{-j2\omega}}$ c) $X_2(\omega) = \dfrac{1 + j\omega}{1 - 2j\omega - 3(j\omega)^2}$



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As we saw in the last lecture, the DTFT (and respectively the CTFT) can also be seen as an extension of the Fourier series. However, every periodic signal is not absolutely summable!



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But we would still like to work with periodic signals in the frequency domain, so what is a reasonable representation for the DTFT of a periodic signal? For example, the DTFT of $x[n] = e^{j\omega_0 n}, \omega_0 \in [0, 2\pi)$.



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$$X_d(\omega) = \begin{cases} \text{Non-zero}, & \omega = \omega_0 + 2\pi k, \ k \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}$$



DTFT of Periodic Signals

Let $x[n] = e^{j\omega_0 n}$:

$$X_d(\omega) = \begin{cases} \text{Non-zero}, & \omega = \omega_0 + 2\pi k,\ k \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}$$



Common DTFT Pairs

Signal $x[n]$	DTFT $X_d(\omega)$
$\delta[n]$	1
$u[n]$	$\frac{1}{1-e^{-j\omega}} + \pi\delta(\omega)$
$a^n u[n]$	$\frac{1}{1-ae^{-j\omega}},\ 0 < \lvert a\rvert < 1$
$e^{j\omega_0 n}$	$2\pi\delta(\omega-\omega_0)$
$\cos(\omega_0 n)$	$\pi\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]$
$\sin(\omega_0 n)$	$-j\pi\left[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)\right]$
$\text{rect}\left(\frac{n-k}{L}\right)$	$\frac{\sin\left(\frac{L\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}e^{-j\omega k}$
$\text{sinc}(Ln)$	$\frac{\pi}{L}\text{rect}\left(\frac{\omega}{2L}\right)$
$\text{sinc}^2(Ln)$	$\frac{\pi}{L}\Delta\left(\frac{\omega}{2L}\right)$

DTFT Properties: 2π Periodicity

The DTFT of any discrete-time signal $x[n]$, where its DTFT exists, is 2π-periodic.

$$X_d(\omega) = X_d(\omega + 2\pi k),\ k \in \mathbb{Z}$$



DTFT Properties: Hermitian Symmetry

If $x[n]$ is real-valued with DTFT $X_d(\omega)$, then $X_d(\omega)$ is *Hermitian symmetric*. This means

$$X_d^*(\omega) = X_d(-\omega)$$

which is equivalent to

$$|X_d(\omega)| = |X_d(-\omega)| \text{ and } \angle X_d(\omega) = -\angle X_d(-\omega).$$

*Note: this is an "if and only if" condition!



Example: Checking Hermitian Symmetry

Suppose the DTFT of a discrete-time signal $x[n]$ is given by

$$X_d(\omega) = \omega^2 e^{j\cos(\omega)}.$$

Determine whether $x[n]$ is real-valued.



Concept Check: Hermitian Symmetry and 2π-periodicity

The DTFT of a real-valued signal $x[n]$ is given by $X_d(\omega)$. Which of the following relationships are true? (Select all that are true)

(a) $X_d(0) = X_d(2\pi)$

(b) $X_d\left(\frac{\pi}{3}\right) = X_d\left(-\frac{\pi}{3}\right)$

(c) $X_d\left(\frac{\pi}{3}\right) = X_d^*\left(-\frac{\pi}{3}\right)$

(d) $X_d\left(\pi\right) = X_d(7\pi)$

(e) $X_d(\frac{2\pi}{3}) = X_d^*(\frac{4\pi}{3})$



Other DTFT Properties

Convolution:

$$x[n] * h[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega)H_d(\omega)$$

Time-shifting:

$$x[n-k] \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega k}X_d(\omega), \ k \in \mathbb{Z}$$

Frequency-shifting and Modulation:

$$e^{j\omega_0 n}x[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega - \omega_0)$$

$$x[n]\cos(\omega_0 n) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}X_d(\omega - \omega_0) + \frac{1}{2}X_d(\omega + \omega_0)$$

Multiplication in time (windowing):

$$x[n]w[n] \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2\pi}X_d(\omega) * W_d(\omega)$$



DTFT Properties Table

Property	Signal	DTFT
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(\omega) + bX_2(\omega)$
Time shifting	$x[n-k]$	$X_d(\omega)e^{-jk\omega}$
Frequency shifting	$e^{j\omega_0 n}x[n]$	$X_d(\omega - \omega_0)$
Modulation	$x[n]\cos(\omega_0 n)$	$\frac{1}{2}X_d(\omega - \omega_0) + \frac{1}{2}X_d(\omega + \omega_0)$
Time reversal	$x[-n]$	$X_d(-\omega)$
Conjugation	$x^*[n]$	$X_d^*(-\omega)$
Differentiation	$nx[n]$	$-j\frac{dX_d(\omega)}{d\omega}$
Convolution	$x[n] * h[n]$	$X_d(\omega)H_d(\omega)$
Windowing	$x[n]w[n]$	$\frac{1}{2\pi}X_d(\omega) * W_d(\omega)$
Hermitian symmetry	$x[n]$ real	$X_d^*(\omega) = X_d(-\omega)$
Parseval's relation	$\sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$	$\frac{1}{2\pi}\int_{2\pi} \lvert X_d(\omega)\rvert^2 d\omega$